

AUTONOMOUS UAV NAVIGATION AND LOCALIZATION IN
GPS/GNSS-DENIED ENVIRONMENTS:
A SYSTEMATIC LITERATURE REVIEW OF
SENSOR-FUSION APPROACHES

Version-1 Full-Text Synthesis

Review window: 2010–2026

Screened-included set: 636 studies

Version-1 full-text corpus: 171 studies (26.9%)

Analysis date: September 2026

Abstract

GPS/GNSS denial caused by indoor occlusion, urban canyons, underground operation, foliage, jamming, or spoofing creates a fundamental localization problem for autonomous unmanned aerial vehicles (UAVs). This systematic literature review examines sensor-fusion approaches for navigation and localization under such conditions. The protocol defines a 2010–2026 review window, a UAV-focused population, multi-sensor interventions, localization and robustness outcomes, and indoor, urban, underground, forest, maritime, and electronic-warfare contexts. The review workflow identified 2,000 raw records, retained 1,719 after deduplication, screened 636 studies as included, and completed machine-assisted full-text extraction for 171 locally available papers. The Version-1 full-text corpus is therefore an intermediate subset, not the final complete review population. Descriptively, the extracted studies are concentrated in visual-inertial odometry, visual-LiDAR-inertial fusion, LiDAR-inertial odometry/SLAM, and radio-based positioning. Evaluation modality counts are 22 real-world-only, 78 simulation-only, and 71 both real-world and simulation studies. Quality appraisal assigns 83 studies to Q-high, 81 to Q-medium, and 7 to Q-low using a 10-point rubric. The evidence supports a strong role for complementary sensing but does not support a pooled accuracy ranking because metrics, units, datasets, and reporting completeness are heterogeneous and many numeric values are not reported in the structured table. The main conclusions are consequently descriptive and should be updated when additional full texts become available.

Keywords: UAV navigation; GNSS-denied; GPS-denied; visual-inertial odometry; LiDAR-inertial odometry; SLAM; sensor fusion; systematic review.

1. Introduction

Reliable position estimation is a prerequisite for autonomous UAV flight. When GPS or another GNSS service is unavailable, degraded, jammed, spoofed, or unreliable because of indoor or built-up environments, a vehicle must construct its state estimate from onboard sensing and prior information. Failure can produce drift, loss of scale or observability, collision risk, mission failure, or unsafe return behavior. The problem is particularly difficult in texture-poor interiors, urban canyons, tunnels, mines, forests, smoke or dust, and adversarial electronic environments.

Sensor fusion is central because no single sensing modality is reliable in all of these conditions. Cameras provide geometric and semantic information but are sensitive to illumination, texture, motion blur, and occlusion. Inertial measurements provide high-rate short-term motion information but drift when not corrected. LiDAR can provide range geometry in low-light or texture-poor scenes, although it adds payload, power, and computational requirements. Radio, UWB, radar, sonar, and map-based information can add complementary constraints but may require infrastructure, environmental assumptions, or additional calibration.

The review protocol addresses a gap in systematic comparison of these configurations for UAV navigation in GPS/GNSS-denied environments. It asks:

RQ1. What sensor-fusion configurations are most commonly used?

RQ2. What performance metrics and accuracy levels are achieved in different environments?

RQ3. What algorithmic approaches and comparative advantages are reported?

RQ4. What limitations and future research directions remain?

This manuscript reports a **Version-1 full-text synthesis**. The canonical screened-included set contains 636 papers, but only 171 corresponding PDFs were locally available for structured full-text extraction at the time of analysis. The conclusions therefore characterize the available full-text subset and must not be generalized without qualification to all 636 included records.

2. Methods

2.1 Review design and protocol: The review follows the repository research protocol (00_scope/research_protocol.md). The population is UAVs, drones, and closely related aerial autonomous systems. The intervention is multi-sensor fusion, including combinations of vision, LiDAR, IMU, radar, UWB, and related aiding sensors. Comparisons include baselines and, where reported, GPS-assisted or alternative navigation conditions. Outcomes include localization accuracy, trajectory error, drift, robustness, computational efficiency, and mission success. The context includes indoor, urban canyon, underground, forest, maritime, jamming, and spoofing environments.

2.2 Search and eligibility: The protocol specifies IEEE Xplore, Scopus, Web of Science, and supplementary Google Scholar searching. Its principal search concepts combine GPS/GNSS denial or indoor/underground navigation with localization, navigation, or SLAM and UAV, drone, or robot terms. The publication window is 2010–2026.

Included studies are English-language, peer-reviewed journal or conference papers that address GPS/GNSS-denied navigation, focus on UAVs/drones/robots, present a sensor-fusion approach, and provide experimental validation or simulation results. The protocol excludes purely theoretical

work without validation, GPS-only navigation, terrestrial-only vehicles, single-sensor-only studies, non-English publications, and work published before 2010.

2.3 Selection and Version-1 corpus construction: The repository records 2,000 raw records, 1,719 records after deduplication, and 636 screened-included studies. The full-text corpus was constructed by matching locally available PDF stems in `05_papers_fulltext/` to study IDs in `screened_included_v2.csv`. This produced 171 matched records, equivalent to 26.9% of the screened-included set. The denominator for full-text coverage is therefore the 636 included studies, not the 1,719 deduplicated records.

2.4 Data extraction: The authoritative table is `02_data_processed/extracted_master_v2.csv`. It contains study identity, authors, year, DOI, venue, full-text status, page count, platform type, sensors, primary method, method category, environment, experiment type, real/simulation status, reported metrics, ATE/RMSE field, application domain, multi-agent status, notes, four quality components, total QA score, QA tier, and citation tier.

Extraction was machine-assisted from PDF text using the repository Phase-7 runner. The resulting labels are suitable for a reproducible descriptive baseline but are not equivalent to independent human coding. A 20-paper stratified validation sample was generated in `08_docs/extraction_validation_sample_v1.csv`; its human-review fields remain to be completed unless subsequently updated.

2.5 Quality appraisal: Each extracted paper receives four components:

Component	Range	Interpretation
Experimental rigor	0–4	Design clarity, validation, ground truth, and repeatability
Reporting completeness	0–3	Metrics, scale, hardware/These and failure reporting
Baseline fairness	0–2	Baseline presence and matched comparisons
Reproducibility	0–1	Code, data, or sufficiently detailed parameters
Total	0–10	Sum of the four components

The tiers are Q-high (7–10), Q-medium (4–6), and Q-low (0–3). A quality tier is a reporting and evidence-quality indicator; it is not proof that the proposed method is superior.

2.6 Synthesis: The synthesis is descriptive. Frequencies and cross-tabulations are reported for methods, sensors, environments, applications, experiment modality, and QA tier. Environment and sensor fields are multi-valued, so their frequency sums can exceed 171. Because the extracted performance values use heterogeneous metrics, units, datasets, and evaluation protocols, no pooled meta-analysis or universal method ranking is attempted.

3. Results

3.1 Selection and corpus profile:

Stage	Records
Raw records	2,000
After deduplication	1,719
Screened-included studies	636
Full-text PDFs locally available and matched	171
Full-text coverage of included set	26.9%

The Version-1 corpus is thus a convenience-of-access subset of the included set. It is appropriate for an interim synthesis and pipeline validation, but not for final prevalence estimates over all included studies.

3.2 Publication years and platforms: The 171 extracted records span 2013–2026. The largest annual group in the current corpus is 2026 (36 records), followed by 2018 (24), 2019 (21), and 2022 and 2023 (19 each). These counts should be interpreted as the temporal distribution of the currently available PDFs, not as the publication trend of the full 636-paper set.

The platform field identifies 95 records as UAV and 76 as multi-platform. Because the latter label can represent papers whose text mentions or evaluates more than one platform, it should not be read as evidence that 76 studies are non-UAV studies.

Sensor field	Papers
IMU	170
3-D LiDAR	93
Stereo camera	87
mmWave radar	48
Monocular camera	46
UWB	46
Sonar	34
Optical flow	30
2-D LiDAR	25
Thermal camera	23
Depth camera	6

3.3 Sensor configurations: These are multi-label problems. IMU appears in nearly all extracted records, while cameras and LiDAR are frequently omitted. The presence of a sensor in a paper does not establish that it was the primary source of accuracy or that it was evaluated under the same conditions as another paper.

3.4 Algorithmic taxonomy:

Primary method	Papers
Visual-LiDAR-inertial fusion	73
Visual-inertial odometry	61
LiDAR-inertial odometry/SLAM	19
Radio-based positioning	9
Thermal-inertial odometry	4
Visual SLAM	3
Hybrid inertial fusion	2

The largest categories combine complementary geometric and inertial information. Visual-inertial methods offer compact, high-rate estimation but remain vulnerable to

visual degradation and accumulated drift. LiDAR-inertial methods provide direct geometric constraints and can be advantageous in texture-poor or low-light settings, at the cost of payload and processing requirements. Radio-based methods can provide global or infrastructure-linked constraints, but their effectiveness depends on anchor placement, propagation, or environmental availability. These are trade-offs represented in the current records, not a universal performance ordering.

3.5 *Environments and applications:*

Environment label	Papers
Adversarial/EW	168
Urban	154
Indoor	144
Underground/tunnel	136
Forest/canopy	89
Maritime/water	66
Mixed/unstructured	1

The unusually high multi-label counts show why environment categories must not be added as mutually exclusive classes. They reflect the labels produced by the current extraction process and require human validation before being used for precise prevalence claims.

Application labels are inspection (107), mapping/surveying (49), search-and-rescue (11), general navigation (3), and defense/tactical (1). Inspection and mapping are therefore the most frequent application labels in the Version-1 table.

3.6 *Evaluation modality and reported metrics:*

Evaluation modality	Papers
Simulation only	78
Real-world only	22
Both real-world and simulation	71

The current corpus contains a substantial simulation component: 78 simulation-only records and 71 records reporting both types of evaluation. This supports a recurring concern about transfer from controlled simulation to physical flight, especially under illumination change, occlusion, vibration, communication loss, and degraded sensing.

The structured metric labels occur as follows: ATE/RMSE (170), success rate (150), computational latency (114), relative drift percentage (110), and RMSE (45). These labels are multi-label. Although ATE/RMSE is frequently marked as a reported metric, the numeric `ate_rmse_m` field is `NOT_REPORTED` or otherwise unavailable for most records. Accordingly, this review does not claim a pooled meter-level accuracy or rank methods by accuracy.

	QA tier	Papers
3.7 <i>Quality appraisal:</i>	Q-high	83
	Q-medium	81
	Q-low	7
	Total	171

Citation tiers are Core (71), Important (93), and Peripheral (7). The Core/Important working subset therefore contains 164 records. The QA distribution indicates that most

currently extracted papers report enough information to receive Q-high or Q-medium status under the repository rubric, but the scores remain machine-assisted and pending sample-level human review.

3.8 Multi-agent systems: The extracted field identifies 85 multi-agent and 86 single-agent records. Multi-agent status should be interpreted cautiously because terminology such as swarm, cooperative, or multi-robot can be detected from text without proving that scalable cooperative localization was experimentally demonstrated. The key unresolved evidence needs are communication reliability, synchronization, relative-observation quality, network scale, and failure recovery.

4. *Discussion*

4.1 RQ1: Sensor-fusion configurations: The Version-1 corpus is dominated by combinations of inertial sensing with vision and/or LiDAR. IMU appears in 170 of 171 records, while 3-D LiDAR and stereo cameras appear in 93 and 87 records, respectively. This pattern is consistent with complementary observability: inertial data provide short-term motion propagation, cameras provide visual constraints, and LiDAR provides range geometry. Radio/UWB, radar, sonar, optical flow, thermal cameras, and depth cameras occur less often but represent targeted responses to specific failure modes or operational constraints.

The frequency pattern answers which configurations are common, but not which configuration is best. A valid comparative answer requires matched datasets, common ground truth, comparable trajectories, and consistent reporting of compute and environmental conditions, which are not uniformly available in the current table.

4.2 RQ2: Metrics and accuracy: The records report a heterogeneous collection of ATE, RMSE, relative drift, success, and latency indicators. The distribution shows substantial attention to trajectory and operational metrics, but the structured numeric field is mostly missing or not reported. Differences in units, sequences, trajectories, flight speed, map scale, ground-truth source, and failure definitions prevent direct numerical pooling. The defensible conclusion is therefore that the field commonly evaluates trajectory error, drift, success, and computation, not that any one method achieves a universal accuracy level.

4.3 RQ3: Algorithmic approaches and comparative advantages: Visual-LiDAR-inertial fusion is the largest extracted method category, followed by visual-inertial odometry. LiDAR-inertial and radio-based approaches form smaller but technically important groups. Visual-inertial systems offer compactness and high-rate updates; LiDAR-inertial systems can strengthen geometric observability where visual texture is poor; radio-based systems can provide external constraints where infrastructure is available. The comparative advantage of each approach is conditional on environment, payload, latency, calibration, and failure mode. The present evidence does not justify declaring a universal winner.

4.4 RQ4: Limitations and future directions: The most important limitations are incomplete full-text availability, heterogeneous evaluation, limited numeric reporting, and the simulation-to-real gap. Future studies should report common benchmark sequences, sensor calibration, synchronization, ground-truth source, trajectory length, runtime, power or payload constraints, failure cases, and matched baseline conditions. Field evaluations should include deliberate degradation, such as lighting change, motion blur, dust, foliage, occlusion, magnetic or radio disturbance, and communication loss. Multi-agent studies should report network scale, bandwidth, latency, packet loss, synchronization, and recovery behavior.

5. Limitations and Threats to Validity

First, only 171 of 636 included papers had local PDFs at the time of extraction. Access and retrieval availability can bias the Version-1 corpus toward recent, open, or institutionally accessible work. Second, the extraction is machine-assisted. Sensor, environment, method, and metric labels may contain false positives or omissions; the 20-paper validation sample must be manually completed before the results are treated as final coded evidence. Third, the studies use incompatible datasets, hardware, trajectories, metrics, and definitions of success, so pooled effect sizes and universal rankings are not appropriate. Fourth, multi-label fields produce counts greater than the number of papers and cannot be interpreted as mutually exclusive distributions. Finally, the 2026 portion of the corpus and the retrieval state are time sensitive and should be regenerated when the search or full-text set changes.

6. Conclusion

This Version-1 full-text synthesis establishes a reproducible baseline for reviewing GPS/GNSS-denied UAV navigation. The available 171 records show a strong concentration of visual-inertial and visual-LiDAR-inertial approaches, frequent use of IMU, camera, and LiDAR sensing, and substantial reliance on simulation or mixed evaluation. They also show that the literature commonly reports trajectory, drift, success, and computational metrics but does not provide a sufficiently uniform numeric basis for pooled accuracy ranking.

The conclusions are intentionally limited to the currently available full-text subset. When additional PDFs arrive, the Phase-7 extraction runner, validation sample, core subset, figures, synthesis matrices, and manuscript must be regenerated. The final review should incorporate the enlarged corpus and completed human validation.

7. Data and Code Availability

The analysis tables and scripts are stored in the repository. The principal files are:

- 02_data_processed/extracted_master_v2.csv
- 02_data_processed/screened_included_v2.csv

- 02_data_processed/screened_included_v2_fulltext.csv
- 02_data_processed/core_papers_v1.csv
- 06_analysis/scripts/16_execute_phase7_extraction.py
- 06_analysis/scripts/06_generate_figures_v2.py
- 06_analysis/scripts/07_generate_synthesis.py
- 08_docs/extraction_validation_sample_v1.csv

Local PDFs are not assumed to be redistributable and may be gitignored or subject to publisher access restrictions. Study-level DOI metadata should be resolved against the master CSV and checked before submission.

8. Figure Captions

Figure 1. Publication trends. Annual distribution of records in the Version-1 full-text corpus; this is not the trend for all 636 included records.

Figure 2. Platform distribution. Extracted platform labels, including multi-platform records.

Figure 3. Environment distribution. Multi-label environment frequencies; categories are not mutually exclusive.

Figure 4. Method evolution. Distribution of extracted primary methods by publication year in the available full-text corpus.

Figure 5. Sensor frequency. Multi-label sensor frequencies, showing the co-occurrence of inertial, camera, LiDAR, radio, radar, sonar, and related modalities.

Figure 6. Application domains. Extracted application-domain labels.

Figure 7. PRISMA-style flow. Review-record counts from raw identification through deduplication, inclusion, and Version-1 full-text extraction.

Figure 8. Method-environment heatmap. Descriptive co-occurrence of primary method categories and environment labels.

Figure 9. Research-maturity radar. Descriptive quality and evaluation dimensions for the available full-text corpus; not a causal or predictive score.

9. Conflicts of Interest and Funding

No conflict-of-interest or funding information was supplied in the repository materials used for this Version-1 synthesis. These statements require author confirmation before submission.

10. References

The repository bibliography (07_manuscript/references.bib) contains verified methodological and benchmark references, including ORB-SLAM3, VINS-Mono, LOAM, FAST-LIO2, LIO-SAM, and the EuRoC dataset. The final manuscript reference list must additionally be generated and checked from DOI metadata for the included study records. No bibliographic details should be added from memory.

Appendix A. Reproducibility and Quality-Control Audit

Check	Result
<code>extracted_master_v2.csv</code> row count = 171	PASS
<code>screened_included_v2.csv</code> row count = 636	PASS
Full-text subset row count = 171	PASS
Full-text coverage = $171/636 = 26.9\%$	PASS
QA totals within 0–10	PASS
QA tiers valid	PASS
QA counts = 83 / 81 / 7	PASS
Experiment counts = 22 / 78 / 71	PASS
Citation tiers = 71 / 93 / 7	PASS
Core + Important = 164	PASS
Human validation sample completed	REVIEW REQUIRED
Complete final-corpus claim justified	NO

Overall status: REVIEW REQUIRED. The numerical Version-1 checks pass, but human validation and additional full-text retrieval remain incomplete. The manuscript is suitable as an intermediate synthesis and reproducibility baseline, not as the final complete-corpus submission.